Practice workbook · Hardware devices: input, output and how they connect

Input and output devices, the role of hardware versus software, and the direction of communication between peripherals and the CPU.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-02

1. Question 5 · part b.i · 3 marks

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Worked-example retry · Attempt part b.i only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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1. Question 5 · part b.i · 3 marks

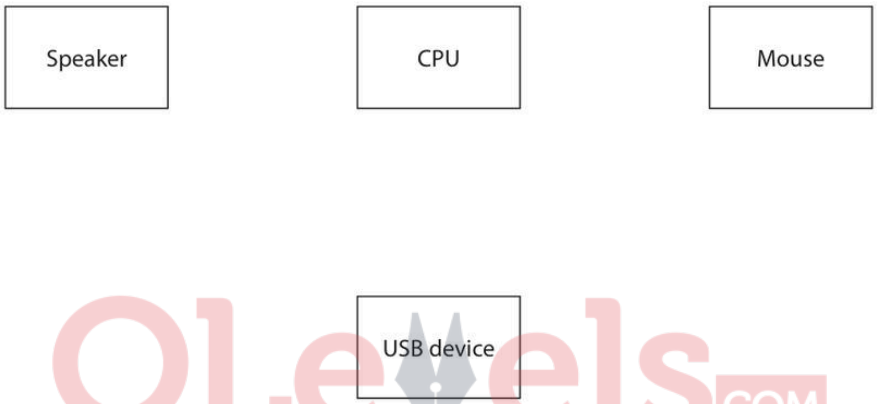
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Worked-example retry · Attempt part b.i only.

(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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Turn over

1. Question 5 · part b.i · 3 marks

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Worked-example retry · Attempt part b.i only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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1. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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2. Question 3 · part a · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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2. Question 3 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part d · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

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3. Question 3 · part d · 2 marks

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Further practice · Attempt part d only.

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(d) Zakir uses many devices.

Complete the table by adding **one** tick (✓) in each row to show whether the device is an input device, an output device or both.

(2)

Device	Input	Output	Both
Headset			
Mouse			
Printer			
Touch screen			

(e) Zakir needs to protect his laptop against malware.

Give **two** types of anti-malware software he could install.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

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Working and answer

4. Question 2 · part a · 2 marks

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Further practice · Attempt part a only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

.....

2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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P 7 8 6 3 7 A 0 5 2 0

4. Question 2 · part a · 2 marks · Working and answer

Working and answer

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